RHOM OUT OF A BOUNDED ABOVE COMPLEX OF PROJECTIVES IS REPRESENTED BY THE TERMWISE HOM COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The cohomology groups of the termwise Hom complex Hom(P, L) compute the Hom groups in the homotopy category when P is bounded above of projectives. invoked at proof lines 18717-18719. Says the cohomology of the candidate complex is correct.
- A quasi-isomorphism between complexes induces a quasi-isomorphism of Hom complexes out of a bounded above complex of projectives. invoked at proof lines 18720-18727. Identifies Hom(P, L) with Hom(P, I) in the derived category.

Source reference. Original equation 0A5X: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17875–17875. Cited at proof line 18717 for the correctness of the cohomology groups of the Hom complex; the formula is used inside the counted span.

Source reference. Original lemma 064B: sources/stacks-chow/tags.txt, source lines 7860–7860. Cited at proof lines 18717-18718 for the computation of Hom groups out of the bounded above projective complex; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $P^\bullet$ be a bounded above complex of projective R-modules. Let $L^\bullet$ be a complex of R-modules. Then $R\mathrm{Hom}_R(P^\bullet, L^\bullet)$ is represented by the complex $\mathrm{Hom}^\bullet(P^\bullet, L^\bullet)$.*

Proof. By (0A5X) and Derived Categories, Lemma 064B the cohomology groups of the complex are "correct". Hence if we choose a quasi-isomorphism $L^\bullet \to I^\bullet$ with $I^\bullet$ a K-injective complex of R-modules then the induced map

$$\mathrm{Hom}^\bullet(P^\bullet, L^\bullet) \longrightarrow \mathrm{Hom}^\bullet(P^\bullet, I^\bullet)$$

is a quasi-isomorphism. As the right hand side is our definition of $R\mathrm{Hom}_R(P^\bullet, L^\bullet)$ we win. □

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